

Yubin Jeon

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RESEARCH INTERESTS

Robot learning for manipulation: how to collect and structure training data, and how to combine high-level planning with low-level control, to solve long-horizon, contact-rich tasks.

- Human-demonstration data collection: datasets from diverse setups such as teleoperation and wearable gloves that go beyond image-action pairs to include object pose, contact points, and contact forces as observations, which are hard to collect in the real world yet essential for more advanced robot learning
- Hierarchical planning and control: a high-level planner such as a VLM produces plans that diverse low-level controllers (VLA, RL, or classical controllers) execute, solving long-horizon, contact-rich tasks beyond the reach of a single policy and scaling to new tasks and robots

EDUCATION

Konkuk University

B.S. in Smart Vehicle Engineering, Institute of Convergence Engineering

Seoul, Republic of Korea

Mar. 2021 – Feb. 2027 (Expected)

- Cumulative GPA: 4.06 / 4.50 Major GPA: 4.08 / 4.50

H-Mobility Class — Robotics Track

Industry-academia program: ROS/Python, kinematics & motion planning, control & dynamics, perception & state estimation

Hyundai Motor Group

Apr. 2026 – Jun. 2026

RESEARCH EXPERIENCE

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

Undergraduate Research Intern, ROBIN Lab

Daegu, Republic of Korea

Jul. 2026 – Sep. 2026

- **VICON (Visual-Inertial-Contact based Hand-Object Tracking)**: Developed a multimodal human-demonstration framework that jointly captures hand motion, object pose, contact locations, and normal-force measurements using a wearable glove and a single RGB-D camera, enabling contact-rich data collection without a motion-capture setup.
 - **Occlusion-Robust Object Tracking**: designed an offline factor-graph refinement that combines visibility-aware object-pose estimates with contact-conditioned hand-motion cues, maintaining trajectories through severe hand occlusion and reducing tracking failures from 50.9% (FoundationPose) to 2.5%.
 - **Contact & Force Annotation**: reconstructed metric object meshes from monocular video and mapped synchronized glove force measurements onto the object surface to recover object-frame contact points and calibrated normal forces.
 - **Robot-Learning Data Pipeline**: built an end-to-end recording and annotation pipeline for synchronized RGB-D, hand-object trajectories, contact, and force signals, designed for force-aware manipulation and VLA training.

Korea Institute of Industrial Technology (KITECH)

Student Researcher, National Engineering R&D Center

Ansan, Republic of Korea

Jan. 2026 – Jun. 2026

- **Skill-wise Expert based Data Generation**: To reduce the high cost of teleoperation data collection, generated VLA fine-tuning data by simulator rollouts with skill-wise experts (teleop demos → Diffusion Policy based BC → Residual RL).
- **VIVA (VLM-Instructed VLA Architecture)**: To overcome the long-horizon limitation of a single VLA model, designed a hierarchical VLM-VLA architecture where Qwen3-VL decomposes commands into skill plans and asynchronously updates $\pi 0.5$'s instructions based on real-time scene understanding. Achieved 38% / 28% / 18% success rates on Easy / Middle / Hard tasks vs. 2% / 6% / 0% (Only VLA baseline).

PUBLICATIONS & PRESENTATIONS

Submitted to IEEE International Conference on Robotics and Automation (ICRA 2027)

Sep. 2026

- **YuBin Jeon***, UiSeong Shin*, HwanChul La*, Jaeseong Kang*, YongSeok Lee†. “VICON: Visual-Inertial-Contact based Hand-Object Tracking for Manipulation Datasets.” (*Under Review*)

KSME 2026 Spring Annual Conference

Apr. 2026

- **Yubin Jeon***, S. Bae, Y.-G. Eun, S.-W. Kim. “VIVA: VLM-Instructed VLA Architecture for Long-Horizon Tasks and Skill-wise Expert based Data Generation Pipeline.” (Poster)

PROJECTS

Real-time Emergency-Stop Failure Prediction | *PyTorch, PyBullet*

Sep. 2025 – Dec. 2025

Course project (Machine Learning, Konkuk University).

- Developed a time-series classifier that predicts vehicle braking failure caused by road condition changes (rain, snow) and triggers an early warning before failure occurs.
- **PyBullet simulation environment**: collected training data by randomizing physical variables (speed, mass, friction, braking timing) and integrated the model’s threshold-based early-warning system into the simulator.
- **Transformer model training and evaluation**: achieved F1 0.89 and Acc 93.4%, selected as the final model with stable Precision of 93%, compared to GRU’s 53%.

Path Planning & Control in Complex Driving Environments | *ROS2, CARLA*

Mar. 2025 – Jun. 2025

Konkuk Dream Semester independent project.

- Built a ROS2 + CARLA autonomous-driving system that navigates safely along an optimal route to the destination across diverse urban scenarios with static and dynamic obstacles and barricaded routes, using coordinated planning and control algorithms.
- **Planning & decision-making module**: designed the ROS2 node architecture and end-to-end pipeline that smoothly switches between algorithms based on driving context using simulator sensor data, and implemented A* global path planning, cubic/B-spline interpolation with curvature-based velocity planning, RRT-based static obstacle avoidance, and D* Lite-based global replanning.

UAV-UGV Cooperative Reconnaissance Platform | *ROS2, PX4, Gazebo*

Mar. 2025 – Jun. 2025

Course project (Autonomous System Platform, Konkuk University).

- Built a cooperative-mission system for a UGV to transport a UAV to a takeoff point for reconnaissance, then meet it at a rendezvous point for landing. **Ranked 1st in course final evaluation.**
- **UGV planning and control**: designed routes and velocity profiles for the UGV to reach the takeoff point without dropping the carried UAV, and quickly return to the rendezvous point for the UAV’s precision landing.
- **UAV ArUco marker global localization**: transformed the camera-frame relative coordinates of detected markers into world-frame global coordinates using the UAV’s pose.

AWARDS & HONORS

President’s List (Rank 1 / 37) — Konkuk University

Spring 2025

- Achieved 4.50 / 4.50 GPA, KU Convergence Engineering Institute.

Seoul RISE Director's Award — 1st Physical AI Hackathon

Feb. 2026

- **Team Lead**, Team Ipalcheongchun. Track: Fine Folding (bimanual cloth folding).
- **Contribution**: Bimanual teleop setup, demo collection, ACT model training, and inference.

Encouragement Award — HL FMA 2025 aMAP Stride Championship

Nov. 2025

- **Team Lead**, Team Ganadi. MORAI + ROS1.
- **Contribution**: GPS-IMU fusion for state estimation, curvature-based velocity planning, custom ROS messages for planning and control.

Special Award — 2025 University Creative Mobility Competition, Autonomous Driving Track

Nov. 2025

- **Planning/Control SW Module Member**, Team KAI.
- **Contribution**: Centroid-graph-based Greedy path planner, control using steering sensitivity adaptive Pure Pursuit algorithm.

Excellence Award — 2025 University Creative Mobility Competition, Creative Technology Track

Nov. 2025

- **Planning/Control SW Module Member**, Team KAI.
- **Contribution**: Delaunay-triangulation based path generation for cone gate driving, control using curvature and velocity adaptive Pure Pursuit algorithm.

Grand Prize — 8th Kookmin University Autonomous Driving Competition

Aug. 2025

- **Team Lead**, Team KAI.
- **Contribution**: Reed-Shepp + Stanley parking system in Preliminary (Gazebo) / lane estimation with YOLO and obstacle distance estimation via camera-LiDAR calibration for overtaking driving in Final (ROS2, real vehicle).

ETRI Director's Award — 4th International Collegiate EV Autonomous Driving Competition

Jul. 2025

- **Team Lead**, Team S.V.E. Basic Autonomous Mobility Race.
- **Contribution**: micro-ROS based Arduino-Jetson data communication pipeline, wall-section driving using 2D LiDAR + VFH algorithm, global pose estimation via motor encoder + IMU odometry for waypoint navigation.

TECHNICAL SKILLS

Programming Languages: Python, C++

Robot Learning: LeRobot, openpi, Isaac Sim, Isaac Lab, PyTorch

Frameworks and Tools: ROS2, ROS1, Gazebo, CARLA, PyBullet, Linux, Git